

St George Girls High School

Trial Higher School Certificate Examination

2007



Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Write your student number on every booklet
- Begin each question in a new booklet
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks (120)

- Attempt Questions 1-10
- All questions are of equal value

Students are advised that this is a Trial Examination only and does not necessarily reflect the content or format of the Higher School Certificate Examination.

Question 1 (12 marks) – Start a new booklet

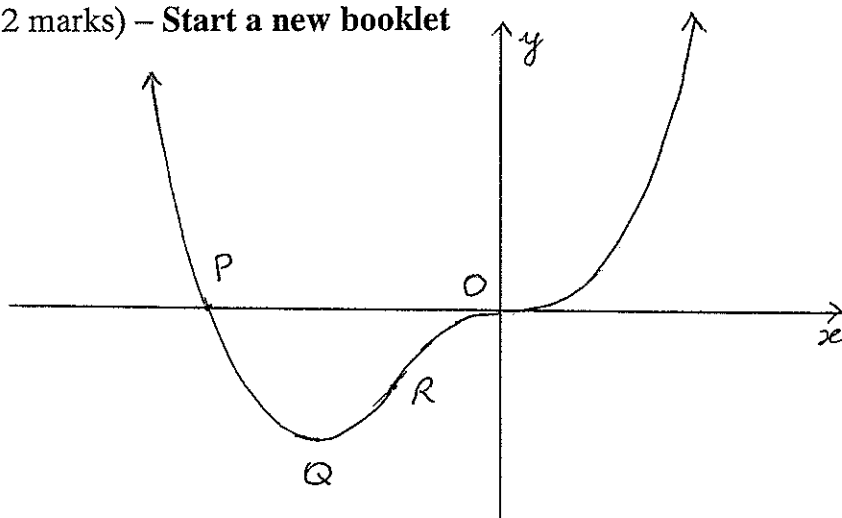
Marks

- a) Find the value of $\frac{27.1}{\sqrt{25.04 \times 57.2}}$ correct to 3 significant figures 2
- b) By rationalizing the denominator, express $\frac{\sqrt{5}}{4 - \sqrt{5}}$ in the form $a + b\sqrt{5}$ 2
- c) Differentiate $\tan \frac{x}{2}$ with respect to x 1
- d) Solve for x : $\frac{x-2}{2} + \frac{x+1}{5} = 2$ 3
- e) Find the primitive function of $3 - \frac{3}{x^2}$ 2
- f) Express $\frac{1-x^{-2}}{1-x^2}$ in its simplest form 2

Question 2 (12 marks) – Start a new booklet

Marks

a)



The diagram shows a sketch of the curve $y = 2x^3 + x^4$. The curve cuts the x -axis at P , has a minimum at Q and a point of inflection at R .

6

(i) Find the coordinates of P .

(ii) Find the coordinates of Q .

(iii) Find the coordinates of R .

b) Solve $|x-1| = 2x-1$

3

c) Consider the parabola with equation $y^2 = 12(1-x)$

3

(i) Find coordinates of the vertex of the parabola.

(ii) Find the coordinates of the focus of the parabola.

Question 3 (12 marks) – **Start a new booklet**

Marks

a) Differentiate the following functions with respect to x

6

(i) $(2 - 3x)^5$

(ii) $x \ln x$

(iii) $\frac{2x}{x^2 - 1}$

b) On the number plane, the points $A(-1, 1)$, $B(4, 3)$, $C(3, 6)$ and $D(-2, 4)$ form a parallelogram.

6

(i) Show that the equation of AB is $2x - 5y + 7 = 0$

(ii) Find the perpendicular distance from the point C to the line AB .

(iii) Hence find the area of the parallelogram $ABCD$

2

Question 4 (12 marks) – Start a new booklet

Marks

a) (i) Evaluate $\int_1^3 e^{2x} dx$

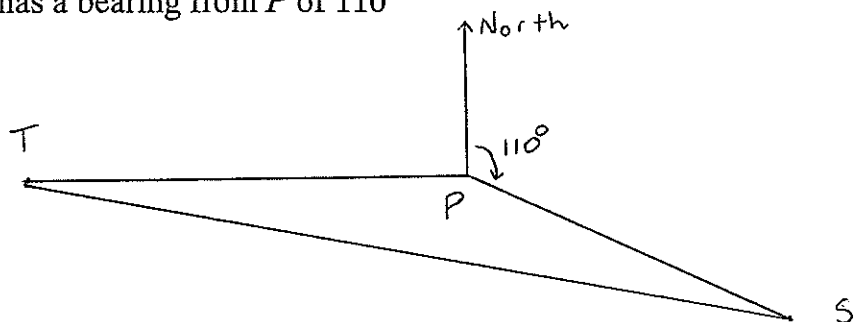
2

(ii) Find $\int \frac{x^2}{x^3 - 4} dx$

2

- b) The diagram shows a point P which is 20km due east of the point T . The point S is 10km from P and has a bearing from P of 110°

4



- (i) Find the length of TS .
- (ii) Find the bearing of T from S .

- c) In a card competition the first prize is \$100, the second prize is \$90, the third prize is \$80 and so on.

4

- (i) Write down the value of the n th prize.
- (ii) If the total prize money is \$550 how many prizes will there be?

Question 5 (12 marks) – Start a new booklet Marks

- a) The table shows the values of a function $f(x)$, for 5 values of x .

x	0	2	4	6	8
$f(x)$	12	25	20	18	8

Use Simpson's rule with these 5 values to estimate $\int_0^8 f(x) dx$ 3

- b) (i) Sketch $y = x^2 + 6$ and $y = 12 - x$ on the same axes. 5

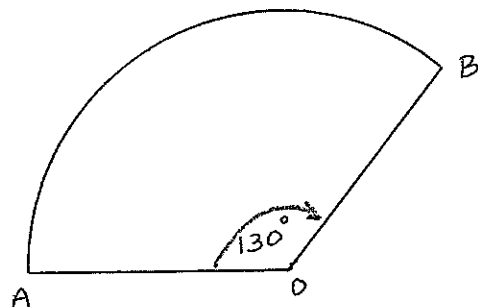
(ii) Find the area in the first quadrant bounded by the y -axis, $y = x^2 + 6$ and $y = 12 - x$

- c) Matthea wishes to invest \$A at the beginning of each month at a compound interest rate of 0.7% per month. How much does she invest each month in order to have \$15 000 saved at the end of the first year. 4

Question 6 (12 marks) – Start a new booklet

Marks

a)



A windscreen wiper 25cm long traces out the above pattern on a wet day.

3

- (i) Express 130° in radians
- (ii) Find the length of the arc AB .
- (iii) Calculate the area of the sector swept out by the wiper.

b) Find the equation of the tangent to the curve $y = e^{2x}$ at the point where $x = 0$

3

c) A particle moves in a straight line such that its distance, x metres, from a fixed point O at any time t seconds is given by:

6

$$x = 4 + 6t - t^3$$

- (i) Find an equation for its velocity after t seconds.
- (ii) At what time does the particle stop?
- (iii) Where is the particle initially?
- (iv) Find the velocity after 2 seconds.
- (v) How far has the particle travelled in the first 2 seconds.

Question 7 (12 marks) – Start a new booklet

Marks

- a) Simplify $\frac{\sin(90 - \theta)}{\cos(180 - \theta)}$ 2
- b) A company donated \$25 000 to charity during 2000. Each subsequent year after 2000 it donated 80% of its previous years donation to charity. 4
- (i) What was the amount donated to charity during 2005?
- (ii) If this arrangement continued indefinitely, what was the maximum total amount the company would donate to charity?
- c) The size of an insect colony is given by the equation $P = 1000e^{kt}$ where P is the population after t days. 6
- (i) What is the initial population?
- (ii) If there are 1200 insects after one day, find the value of k , correct to 2 decimal places.
- (iii) When will the colony double its initial population (answer correct to the nearest day)
- (iv) At what rate is the population growing after 2 days?

Question 8 (12 marks) – Start a new booklet

Marks

- a) (i) Write down the discriminant of $x^2 - 3kx + 9k$ 3
- (ii) For what values of k is $x^2 - 3kx + 9k$ always positive?
- b) If α and β are the roots of $\frac{1}{x} = x + p$, find in terms of p 4
- (i) $\alpha + \beta$
- (ii) $\alpha\beta$
- (iii) $\frac{1}{\alpha} + \frac{1}{\beta}$
- c) Catherine had a full drink bottle containing 500ml of Gatorade. She drank from it so that the volume, V millilitres, of Gatorade in the bottle changed at a rate given by $\frac{dV}{dt} = \left(\frac{2}{5}t - 20\right)$ millilitres per second. 5
- (i) Find a formula for V .
- (ii) Show that it took Catherine 50 seconds to drink the entire contents of the bottle.
- (iii) How long to the nearest second, did it take Catherine to drink half the contents of the bottle?

Question 9 (12 marks) – Start a new booklet

Marks

- a) A and B are the points $(-2, 0)$ and $(2, -1)$ respectively on the number plane. The point P has coordinates (x, y) and given that $\angle APB = 90^\circ$ show that the locus of P is the curve whose equation is $x^2 + y^2 + y = 4$ 4
- b) Consider the function $y = \sin x + \cos x$ in the domain $0 \leq x \leq 2\pi$ 8
- (i) Find $\frac{dy}{dx}$ (
- (ii) Find the maximum and minimum values of $\sin x + \cos x$ in the given domain.
- (iii) Show that the curve cuts the x -axis at $x = \frac{3\pi}{4}$ and $x = \frac{7\pi}{4}$
- (iv) Hence sketch the curve of $y = \sin x + \cos x$ in the domain $0 \leq x \leq 2\pi$ (

Question 10 (12 marks) – **Start a new booklet**

Marks

- a) (i) Sketch the curve $y = \tan x$, $0 \leq x \leq \frac{\pi}{2}$ 4
- (ii) The area under the curve $y = \tan x$ between $x = 0$ and $x = \frac{\pi}{3}$ is rotated about the x -axis. Find the volume of the solid formed.
- b) Emma inherited one million dollars on the 1st of January 2000. She decided to deposit the whole amount into an account which paid 6% p.a. The interest was calculated and paid half-yearly. Emma decided that she would make an annual withdrawal of \$75 000, starting on the 1st of January 2001. 8
- (i) Write down an expression for the amount in Emma's account immediately after the first withdrawal.
- (ii) Show that the amount in Emma's account immediately after her third withdrawal is given by $10^6 \times (1.03)^6 - 75\,000(1 + 1.03^2 + 1.03^4)$
- (iii) At this rate of withdrawal how many years will the money last?

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

SOLUTIONS T.A.S.C. - Mathematics 2007

Q1

(a) 0.0947 (3 sig fig)
($0.09467 \dots$)

(b)

$$\frac{\sqrt{5}}{4-\sqrt{5}} \times \frac{4+\sqrt{5}}{4+\sqrt{5}}$$

$$= \frac{4\sqrt{5} + 5}{16-5}$$

$$= \frac{5}{11} + \frac{4}{11}\sqrt{5}$$

(c) $\frac{d}{dx} \left(\tan \frac{x}{2} \right) = \sec^2 \frac{x}{2} \times \frac{1}{2}$
 $= \frac{1}{2} \sec^2 \frac{x}{2}$

(d) $(5x-10) + (2x+2) = 20$

$$7x = 28$$

$$x = 4$$

(e) $\int (3 - 3x^{-2}) dx$
 $= 3x - \frac{3x^{-1}}{-1} + C$
 $= 3x + \frac{3}{x} + C$

(f) $\frac{1 - \frac{1}{x^2}}{1 - x^2} = \frac{x^2 - 1}{\cancel{x^2} \cancel{1 - x^2}}$

$$= - \frac{\cancel{(1-x^2)}}{\cancel{x^2} \cancel{(1-x^2)}}$$

$$= - \frac{1}{x^2}$$

Q 2 (a) (i) let $x^3(2+x)=0$
 then $x=0$
 so $P(-2,0)$ & $O(0,0)$

(ii) $\frac{dy}{dx} = 6x^2 + 4x^3$

let $6x^2 + 4x^3 = 0$

$2x^2(3+2x)=0$

Start at $x = -\frac{3}{2}$, $x=0$

x	-2	$-\frac{3}{2}$	-1	0	1
y'	-8	0	2	0	10

Min at $x = -\frac{3}{2}$

$\therefore Q\left(-\frac{3}{2}, \left(-\frac{54}{8} + \frac{81}{16}\right)\right)$

$Q\left(-\frac{3}{2}, -\frac{27}{16}\right)$

(iii) let $\frac{d^2y}{dx^2} = 0$

then $12x + 12x^2 = 0$

$12x(1+x)=0$

at $x=-1$ & $x=0$

x	-2	-1	$-\frac{1}{2}$	0	1
y	24	0	-3	0	24

Inflection at $x=-1$

$\therefore R(-1, -1)$

(b) (i) $x-1=2x-1$

$x=0$
 Check $|0-1| \neq 1$
 not a soln

$-(x-1)=2x-1$

$3x=2$

$x=\frac{2}{3}$

Check $\left|\frac{2}{3}-1\right| = \frac{1}{3}$
 $= \frac{4}{3} - 1$ ✓

$x=\frac{2}{3}$ is a soln

(c) (i) Vertex $(1,0)$

(ii) $(y-0)^2 = -12(x-1)$
 $= -4 \times 3(x-1)$

$a=3$

Focus $(-2,0)$

Q3

$$(a) (i) \frac{d}{dx} [(2-3x)^5] = 5(2-3x)^4 \times -3 \\ = -15(2-3x)^4$$

$$(ii) \frac{d}{dx} [x \ln x] = \ln x \times 1 + x \times \frac{1}{x} \\ = \ln x + 1$$

$$(iii) \frac{d}{dx} \left[\frac{2x}{x^2-1} \right] = \frac{(x^2-1) \times 2 - 2x \times 2x}{(x^2-1)^2} \\ = \frac{2x^2 - 2 - 4x^2}{(x^2-1)^2} \\ = \frac{-2(1+x^2)}{(x^2-1)^2}$$

$$(b) (i) m_{AB} = \frac{3-1}{4-1} \\ = \frac{2}{3}$$

$$\therefore \text{Eqn. } (y-1) = \frac{2}{3}(x-1) \\ 3y-3 = 2x+2$$

$$2x - 3y + 7 = 0$$

$$(ii) d = \frac{|2 \times 3 + -3 \times 6 + 7|}{\sqrt{2^2 + (-3)^2}} \\ = \frac{|-17|}{\sqrt{29}} \\ = \frac{17}{\sqrt{29}}$$

$$\therefore \text{Area} = bh$$

$$= \sqrt{29} \times \frac{17}{\sqrt{29}}$$

$$(iii) d_{AO} = \sqrt{(4-1)^2 + (3-1)^2} \\ = \sqrt{29}$$

$$= 17 \text{ sq. units.}$$

Q4

$$\textcircled{a} \text{ (i) } \int_1^3 e^{2x} dx = \left[\frac{1}{2} e^{2x} \right]_1^3$$

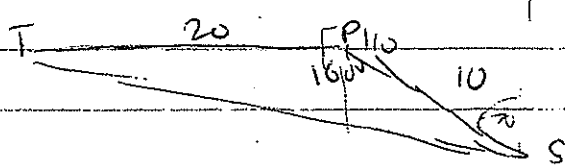
$$= \frac{1}{2} (e^6 - e^2)$$

$$= \frac{e^2}{2} (e^4 - 1)$$

$$\text{(ii) } \int \frac{x^2}{x^3-4} dx = \frac{1}{3} \int \frac{3x^2}{x^3-4} dx$$

$$= \frac{1}{3} \ln(x^3-4) + C$$

b)



$$\text{(i) } TS^2 = 20^2 + 10^2 - 2 \times 20 \times 10 \times \cos 160$$

$$TS^2 = \dots 875.877$$

$$TS = 29.6 \text{ km (1 dec)}.$$

$$\text{(ii) } \frac{\sin S}{20} = \frac{\sin 160}{\sqrt{875.877}}$$

$$\therefore S = 13^\circ 35'$$

So Bearing of T from S

$$= (360 - [70 + 13^\circ 35'])$$

$$= 276^\circ 25'$$

$$\textcircled{c} \text{ (i) } 100, 90, 80, \dots (110 - 10n) \quad [100 + (n-1) \times -10]$$

$$\textcircled{ii} \quad S_n = 550$$

$$\therefore 550 = \frac{n}{2} [2a + (n-1)x - 10]$$

$$550 = \frac{n}{2} [200 - 10n + 10]$$

$$= \frac{n}{2} [210 - 10n]$$

$$1100 = 210n - 10n^2$$

$$\therefore n^2 - 21n + 110 = 0$$

$$(n-10)(n-11) = 0$$

$$n = 10, n = 11$$

?

There are two solutions mathematically.

But, practically we do not award a prize of \$0.

 $\therefore$ Ten prizes given.

$$T_{10} = 100 + (10-1) \times -10$$

$$= 10$$

$$T_{11} = 100 + (11-1) \times -10$$

$$= 0$$

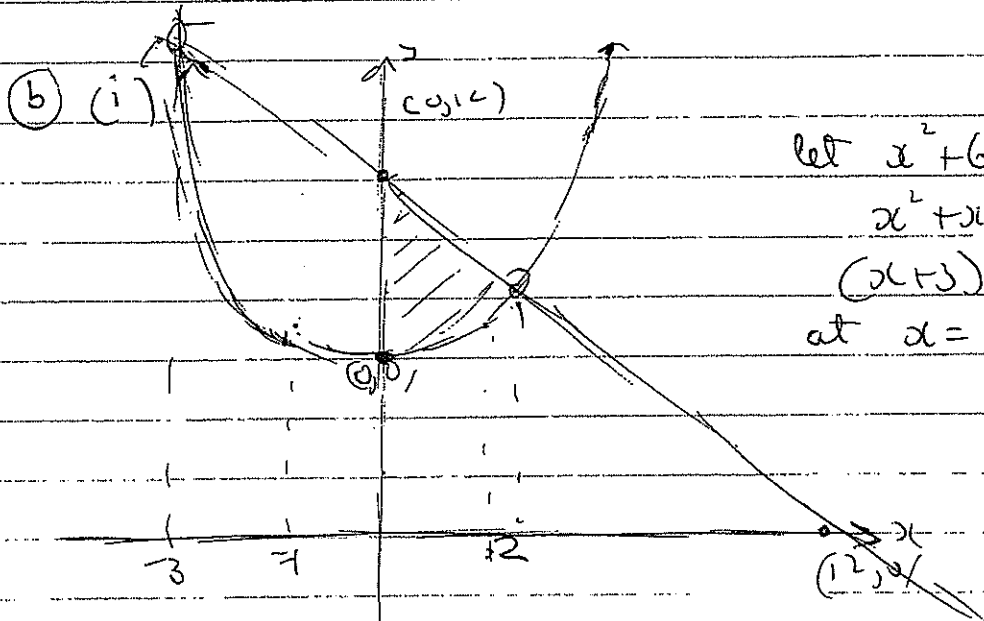
Q5

$$(a) A = \frac{2}{3} \left\{ 12 + 4 \times 25 + 20 \right\} + \frac{2}{3} \left\{ 20 + 4 \times 18 + 8 \right\}$$

$$= \frac{2}{3} \left\{ 224 \right\}$$

$$= \frac{448}{3} \text{ m}^2$$

$$\approx 154 \frac{2}{3} \approx 154.7$$



$$\text{let } x^2 + 6 = 12 - x$$

$$x^2 + x - 6 = 0$$

$$(x+3)(x-2) = 0$$

$$\text{at } x = -3, 2$$

$$(ii) \text{ Area} = \int_{-3}^2 \left[(12-x) - (x^2+6) \right] dx$$

$$= \int_{-3}^2 (6-x-x^2) dx$$

$$= \left[6x - \frac{x^2}{2} - \frac{x^3}{3} \right]_{-3}^2$$

$$= \left(12 - 2 - \frac{8}{3} \right) - 0$$

$$= 7 \frac{1}{3} \text{ m}^2$$

$$(c) \text{ Total} = A(1+0.007)^{12} + A(1+0.007)^{11}$$

$$15000 = A(0.007) \left[1 + 1.007 + \dots + 1.007^{11} \right]$$

$$= A(0.007) \frac{(1.007^{12} - 1)}{(1.007 - 1)} \text{ G.S.}$$

$$A = \frac{15000 \times 0.007}{1.007(1.007^{12} - 1)}$$

$$A = \$1194.24 \text{ needs to be invested}$$

Q6

(a) (i) $130 \times \frac{\pi}{180} = \frac{13\pi}{18}$

(ii) $l = 25 \times \frac{13\pi}{18}$

$= \frac{325\pi}{18} \text{ cm} [56.7 \text{ cm}]$

(iii) $A = \frac{1}{2} \times 25^2 \times \frac{13\pi}{18}$

$= \frac{8125\pi}{36} \text{ cm}^2 [709.5 \text{ cm}^2]$

(b) $\frac{dy}{dx} = e^{2x} \times 2$
 $\frac{dy}{dx} = 2e^{2x}$

at $x=0$, $y=e^0$
 $y=1$

Gradient of tangent

$(0, 1)$

at $x=0$

$m = 2 \times e^0$
 $= 2$

Equation: $y-1 = 2(x-0)$
 $y = 2x+1$

(c) (i) $v = \frac{dx}{dt} = 6-3t^2$

(ii) Stationary when $v=0$, $6-3t^2=0$
 $3(2-t^2)=0$

$t=\sqrt{2} [t>0]$

Stops after $t=\sqrt{2}$ seconds

(iii) Initially $t=0$, $x=4+6 \times 0 - 0^3$
 $x=4$

Initially 4 m to right of origin

(iv) Let $t=2$, $v=6-3 \times 2^2$
 $v=-6 \text{ m/sec.}$

(v) Total distance = $\int_0^{\sqrt{2}} (6-3t^2) dt + \left| \int_{\sqrt{2}}^2 (6-3t^2) dt \right|$
 $= \left[6t - t^3 \right]_0^{\sqrt{2}} + \left| \left[6t - t^3 \right]_{\sqrt{2}}^2 \right|$
 $= (6\sqrt{2} - 2\sqrt{2}) + \left| (12-8) - (6\sqrt{2} - 2\sqrt{2}) \right|$
 $= 4\sqrt{2} + 4\sqrt{2} - 4$
 $= (8\sqrt{2} - 4) \text{ m.}$

* 7.3 m travelled.

Q7

$$\begin{aligned} \textcircled{a} \quad \frac{\sin(90-\theta)}{\cos(180-\theta)} &= \frac{\cos \theta}{-\cos \theta} \\ &= -1 \end{aligned}$$

$$\begin{aligned} \textcircled{b} \quad \textcircled{i} \quad \text{Geometric series, } a &= 25000 \quad r = .8 \quad n = 5 \\ T_5 &= ar^{n-1} \\ &= 25000 \times .8^5 \\ &= 8192 \end{aligned}$$

\$8192 donated in 2005

$$\textcircled{ii} \quad |r| < 1 \text{ gives infinite sum, } |.8| < 1$$

$$S_{\infty} = \frac{25000}{1 - .8}$$

= \$125000 is max. total

$$\begin{aligned} \textcircled{c} \quad \textcircled{i} \quad \text{let } t=0, \quad P &= 1000 e^{k \times 0} \\ P &= 1000 \text{ initial population} \end{aligned}$$

$$\begin{aligned} \textcircled{ii} \quad \text{let } t=1, \quad P &= 1200 \\ 1200 &= 1000 e^{k \times 1} \\ 1.2 &= e^k \end{aligned}$$

$$\ln(1.2) = k \quad \therefore k = 0.18 \text{ [2 dec pl]}$$

$$\textcircled{iii} \quad \text{Find } t \text{ when } P = 2000$$

$$2000 = 1000 e^{kt}$$

$$2 = e^{kt}$$

$$\ln 2 = kt \quad \Rightarrow t = \frac{\ln 2}{\ln(1.2)} = \frac{.693}{.18} = 3.8$$

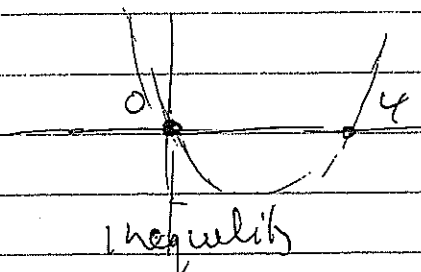
approx 4 days.

$$\begin{aligned} \textcircled{iv} \quad \frac{dP}{dt} &= 1000 k e^{kt} \\ &= k \times 1000 e^{kt} \end{aligned}$$

$$\text{When } t=2, \quad \frac{dP}{dt} = 262.5 \text{ approx 262 insects per day.}$$

Q8 a) (i) $\Delta = (-3k)^2 - 4 \times 1 \times 9k$
 $= 9k^2 - 36k$

(ii) Positive definite $a > 0$, $\Delta > 0$ concave up
 $\Delta < 0$, $\Rightarrow 9k(k-4) < 0$



From graph.

$0 < k < 4$ is

soln

(b) (i) $\alpha + \beta = -\frac{b}{a}$
 $= -\frac{p}{1}$

for $1 = x^2 + px$
 $x^2 + px - 1 = 0$

(ii) $\alpha\beta = \frac{c}{a}$
 $= -1$

(iii) $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\beta + \alpha}{\alpha\beta}$
 $= \frac{-p}{-1}$

$= p$

(c) (c) $v = \int \left(\frac{2}{5}t - 20 \right) dt$

$= \frac{2 \times \frac{1}{2} t^2 - 20t + C$

$v = \frac{1}{5}t^2 - 20t + C$ at $t=0$, $v=500$

$\therefore v = \frac{1}{5}t^2 - 20t + 500$

(ii) Let $v=0$, $\frac{1}{5}t^2 - 20t + 500 = 0$
 $t^2 - 100t + 2500 = 0$
 $(t-50)^2 = 0$

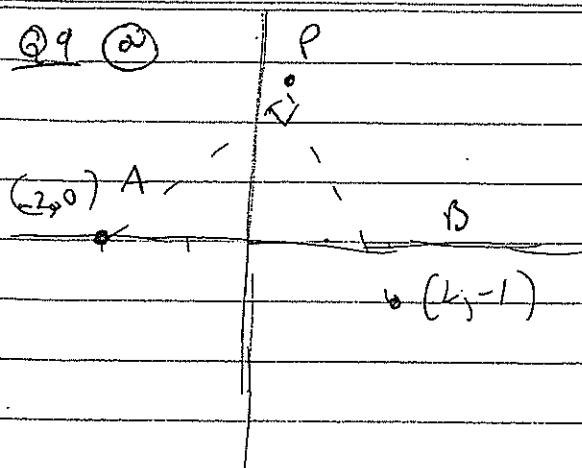
Empty at $t=50$, zero of quadratic

(iii) Let $v=250$, $\frac{1}{5}t^2 - 20t + 500 = 250$
 $t^2 - 100t + 1250 = 0$

$t = 100 \pm \sqrt{100^2 - 4 \times 1 \times 1250}$
 $= 14.6 \text{ sec.} \quad \text{or} \quad 85.4$

Since empty after 50 seconds
it took approx 15 seconds.

Q9 (2)



$$m_{AP} \times m_{BP} = -1$$

[Perpendicular lines $m_1 \times m_2 = -1$]

$$\therefore \frac{y-0}{x-2} \times \frac{y-1}{x-2} = -1$$

$$y(y+1) = -(x+2)(x-2)$$

$$y^2 + y = -(x^2 - 4)$$

$$\therefore x^2 - 4 + y^2 + y = 0$$

$$\text{or } x^2 + y^2 + y = 4$$

(b) (i) $\frac{dy}{dx} = \cos x - \sin x$

(ii) let $\frac{dy}{dx} = 0$, $\cos x - \sin x = 0$

$$\sin x = \cos x$$

$$\tan x = 1$$

[Let $x = \frac{\pi}{4}, \frac{3\pi}{4}$
not $\frac{5\pi}{4}$]

1st & 3rd Q

$$\frac{\pi}{4}$$

$$\frac{5\pi}{4}$$

x	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$
y'	0.5	0	-0.4	-1	0	1.4

Local max

at $(\frac{\pi}{4}, \frac{2}{\sqrt{2}})$

Local Min

at $(\frac{5\pi}{4}, -\frac{2}{\sqrt{2}})$

or $\frac{d^2y}{dx^2} = -\sin x - \cos x$
 $= -(\sin x + \cos x)$

at $x = \frac{\pi}{4}$, $\frac{d^2y}{dx^2} = -(\frac{2}{\sqrt{2}}) < 0$

Concave
down

$x = \frac{5\pi}{4}$, $\frac{d^2y}{dx^2} = -(-\frac{2}{\sqrt{2}}) > 0$

Concave
up

Max value $\frac{2}{\sqrt{2}} \left(\frac{\sqrt{2}}{-1} \right)$; Min value $-\sqrt{2}$

(iii) Solve $\sin x + \cos x = 0$

$$\sin x = -\cos x$$

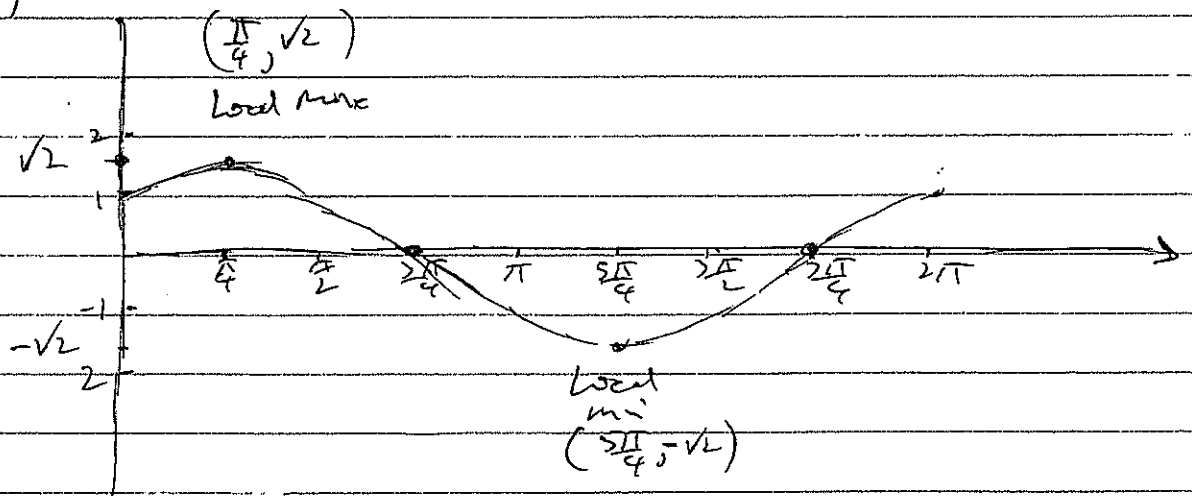
$$\tan x = -1$$

[$\text{clear } \frac{\pi}{2}, \frac{3\pi}{2}$
not sol]

Relevant angle, $\frac{\pi}{4}$ $\frac{2\pi}{4}$ & $\frac{4\pi}{4}$ $\frac{3\pi}{4}$ $\frac{7\pi}{4}$

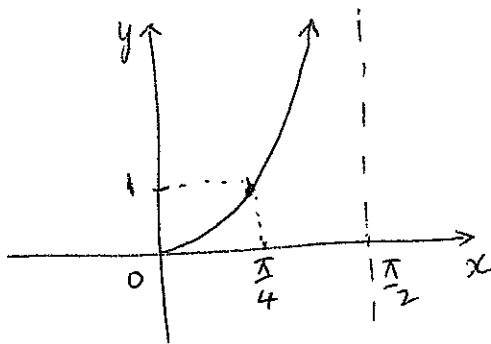
$y=0$ where $x = \frac{3\pi}{4}$, $x = \frac{7\pi}{4}$

(iv)



Q 10.

a) i)



$$\begin{aligned}
 \text{ii) } V &= \pi \int_0^{\pi/3} \tan^2 x \, dx & \frac{S^2}{C^2} + \frac{C^2}{C^2} &= \frac{1}{C^2} \\
 &= \pi \int_0^{\pi/3} (\sec^2 x - 1) \, dx \\
 &= \pi \left[\tan x - x \right]_0^{\pi/3} \\
 &= \pi \left[\left(\tan \frac{\pi}{3} - \frac{\pi}{3} \right) - (\tan 0 - 0) \right] \\
 \text{Vol.} &= \pi \left(\sqrt{3} - \frac{\pi}{3} \right) \text{ m}^3
 \end{aligned}$$

b) i) after 6 mths she has

$$10^6 \times 1.03$$

after 12 mths she has

$$10^6 \times 1.03^2$$

Let A_1 be amt after 1st withdrawal.

$$A_1 = 10^6 \times 1.03^2 - 75000$$

ii) After another 6 mths she has

$$(10^6 \times 1.03^2 - 75000) \times 1.03$$

$$\begin{aligned}
 \therefore A_2 &= (10^6 \times 1.03^2 - 75000) \times 1.03^2 - 75000 \\
 &= 10^6 \times 1.03^4 - 75000 \times 1.03^2 - 75000
 \end{aligned}$$

$$\begin{aligned}
 \text{and } A_3 &= A_2 \times 1.03^2 - 75000 \\
 &= (10^6 \times 1.03^4 - 75000 \times 1.03^2 - 75000) \times 1.03^2 - 75000 \\
 &= 10^6 \times 1.03^6 - 75000 (1.03^4 + 1.03^2 + 1)
 \end{aligned}$$

iii) Following this pattern A_n = amt in account after n withdrawals.

$$A_n = 10^6 \times 1.03^{2n} - 75000 (1 + 1.03^2 + \dots + 1.03^{2n-2})$$

$$a = 1, r = 1.03^2, n \text{ terms}$$

$$A_n = 10^6 \times 1.03^{2n} - 75000 \left(\frac{1.03^{2n} - 1}{1.03^2 - 1} \right)$$

Money will run out when $A_n = 0$.

$$0 = 10^6 \times 1.03^{2n} (1.03^2 - 1) - 75000 \times 1.03^{2n} + 75000$$

$$1.03^{2n} (75000 - 10^6 \times (1.03^2 - 1)) = 75000$$

$$2n \log 1.03 = \log \frac{75000}{75000 - 10^6 (1.03^2 - 1)}$$

$$n = 28.27 \text{ yrs}$$

or 28 yrs 3 mths.

∴ It will take about 28 yrs to run out.